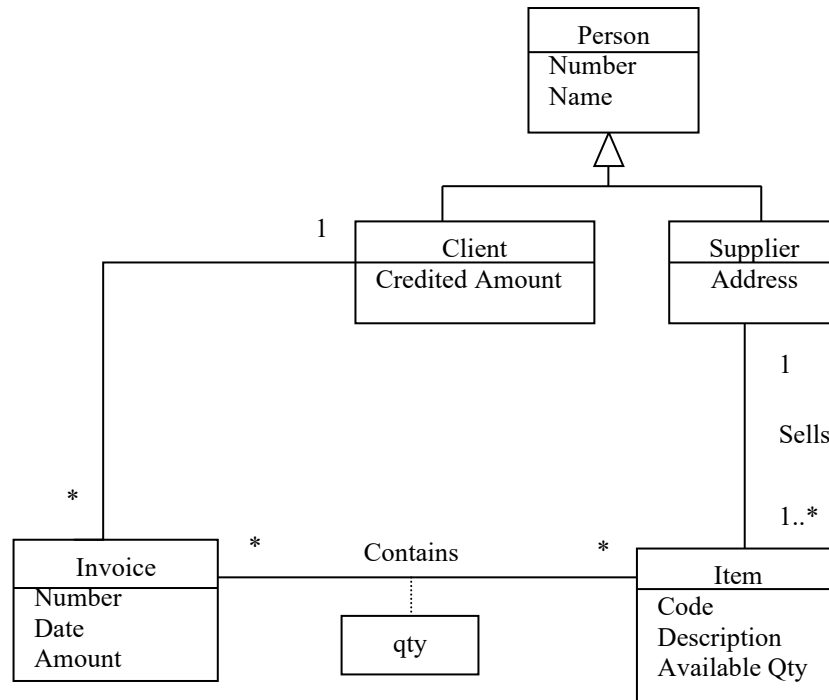


Treat all the following questions:

1. Use the ODL language to transform the following class diagram into object database schema. (5 pts)



2. Given the following relational database: (8 pts)

Employees

EmpNo	EmpName	Salary
E1	Federer	20000
E2	Klum	10000
E3	Spears	12000
E4	Montoya	6000
E5	Beckham	15000
E6	Lopez	18000

ProjectAssignments

EmpNo	ProjNo	MonthsSpent
E1	P1	3
E2	P2	4
E2	P4	2
E3	P3	8
E4	P1	6
E5	P2	9
E6	P4	4.5

Projects

ProjNo	ProjName	Location	Budget
P1	Training	Geneva	98000
P2	Advertising	Geneva	72000
P3	Management	Munich	150000
P4	Customer Support	Zurich	120000

Typical queries from applications are:

At the controlling department located in Geneva:

A1

```

SELECT P.ProjNo, P.Budget, PA.EmpNo, PA.MonthsSpent
FROM Projects P, ProjectAssignments PA
WHERE P.ProjNo = PA.ProjNo AND P.Budget < 110000
    
```

A2

```

SELECT PA.EmpNo, PA.ProjNo, PA.MonthsSpent
FROM Employees E, ProjectAssignments PA
WHERE E.EmpNo = PA.EmpNo AND E.Salary > 11500
    
```

At the human resource department located in Munich:

A3

```

SELECT * FROM Employees E
    
```

A4

```

SELECT E.*, PA.MonthsSpent
FROM Employees E, ProjectAssignments PA
WHERE E.EmpNo = PA.EmpNo
    
```

At the sales department located in Zurich:

A5

```

SELECT * FROM Projects
WHERE Location != "Geneva"
    
```

a) What are the corresponding fragments of the two relations Employees and Projects?

b) How would you fragment horizontally the relation Project Assignments? Give a short discussion of your decision.

3. Given the following relations: (7 pts)

Activity (ActivityName, duration)

Employee (EmpName, JobName)

Job (JobName, ActivityName)

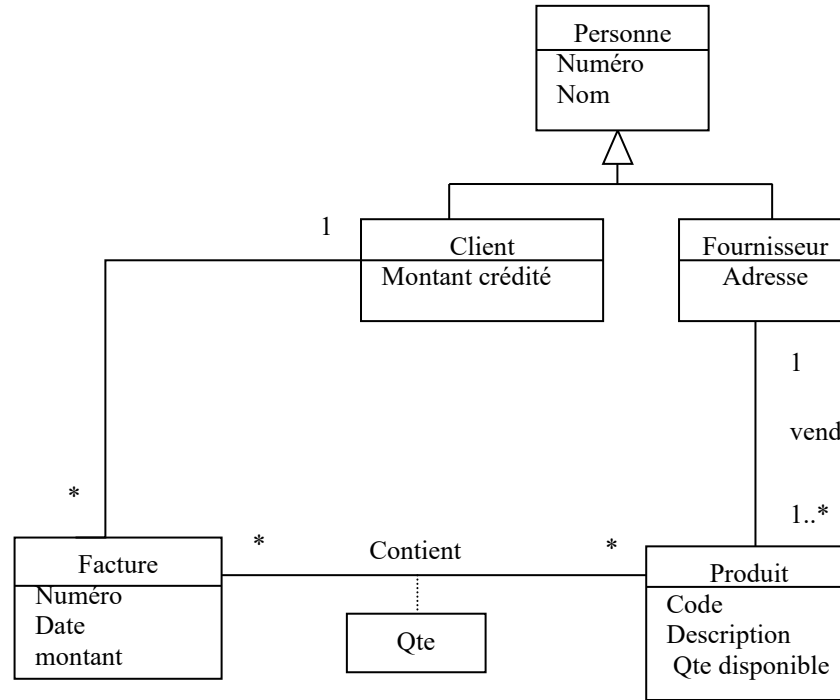
Chief (ChiefName, JobName)

Show the following queries in Datalog:

- a. Who are the employees who work on a job containing the activity “Coding”?
 - b. What are the activities of the Job “Development”?
 - c. Who is the chief of the employee “Fouad”?
 - d. What are the jobs directed by the chief “Tarek”?
 - e. Who are the employees who work on the activities "Coding" and "Testing"?
-

Traitez toutes les questions suivantes:

1. Utilisez le langage ODL pour transformer le diagramme de classes suivant en un schéma de base de données objet. (5 pts)



2. Soit la base de données relationnelle suivante: (8 pts)

Employees

EmpNo	EmpName	Salary
E1	Federer	20000
E2	Klum	10000
E3	Spears	12000
E4	Montoya	6000
E5	Beckham	15000
E6	Lopez	18000

ProjectAssignments

EmpNo	ProjNo	MonthsSpent
E1	P1	3
E2	P2	4
E2	P4	2
E3	P3	8
E4	P1	6
E5	P2	9
E6	P4	4.5

Projects

ProjNo	ProjName	Location	Budget
P1	Training	Geneva	98000
P2	Advertising	Geneva	72000
P3	Management	Munich	150000
P4	Customer Support	Zurich	120000

Les requêtes typiques depuis les applications sont:

Au département de contrôle situé à Genève:

A1

```

SELECT P.ProjNo, P.Budget, PA.EmpNo, PA.MonthsSpent
FROM Projects P, ProjectAssignments PA
WHERE P.ProjNo = PA.ProjNo AND P.Budget < 110000
  
```

A2

```

SELECT PA.EmpNo, PA.ProjNo, PA.MonthsSpent
FROM Employees E, ProjectAssignments PA
WHERE E.EmpNo = PA.EmpNo AND E.Salary > 11500
  
```

Au département de ressources humaines situé à Munich:

A3

```

SELECT * FROM Employees E
  
```

A4

```

SELECT E.*, PA.MonthsSpent
FROM Employees E, ProjectAssignments PA
WHERE E.EmpNo = PA.EmpNo
  
```

Au département de vente situé à Zurich:

A5

```

SELECT * FROM Projects
WHERE Location != "Geneva"
  
```

a) Quels sont les fragments correspondants aux deux relations « Employees » et « Projects »?

b) Comment fragmentez-vous horizontalement la relation ProjectAssignments? Donnez une brève discussion de votre décision.

3. Étant donné les relations suivantes: (7 pts)

Activité (NomActivité, durée)

Employé (NomEmp, NomTravail)

Travail (NomTravail, NomActivité)

Chef (NomChef, NomTravail)

Affichez les requêtes suivantes avec Datalog:

- Qui sont les employés qui travaillent sur un travail contenant l'activité "Codage"?
 - Quelles sont les activités du travail de "développement"?
 - Qui est le chef de l'employé "Fouad"?
 - Quels sont les travaux dirigés par le chef "Tarek"?
 - Qui sont les employés qui travaillent sur les activités «Codage» et «Test»?
-